

MAT1001 Final Examination

Tuesday, December 20, 2022

Time: 6:30 - 9:30 PM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The total score of this examination is **148**.*
3. *There are **15** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Answer all questions on the **answer book**.*
6. *Show your intermediate steps **except Questions 1, 2, and 3** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *The symbols $\arcsin$, $\arctan$, etc., denote the inverse trigonometric functions $\sin^{-1}$, $\tan^{-1}$, etc. .*

MAT1001 Final Examination Questions

1. [12] True or False (in general)? No explanation is required.

For parts (i) to (iv), the functions $f(x)$ and $g(x)$ are assumed to be positive for sufficiently large x , and we consider $x \rightarrow \infty$.

- (i) If $f = o(g)$, then $f = O(g)$. True
- (ii) If f and g grow at the same rate, then $f = O(g)$ and $g = O(f)$. True
- (iii) The function $\log_2$ grows faster than $\log_{10}$. False
- (iv) $\ln(\ln(x)) = O(\ln(x))$. True
- (v) If f and g are both increasing and differentiable on $(-\infty, \infty)$, then the composite function $f \circ g$ is also increasing and differentiable on $(-\infty, \infty)$. True
- (vi) The function $f(x) = x - \sin x$ is increasing on $(-\infty, \infty)$. True
1 - cos X

2. [15] Short Questions. No explanation is required.

- (i) The equation $x^3 - 3y + 1 = e^y$ determines y as a function of x implicitly. Find dy/dx at $x = 0$. (Note that $(x, y) = (0, 0)$ satisfies the equation.)
3x^2 - 3 dy/dx = e^y dy/dx 3x^2 = dy/dx = 0

- (ii) Fill in the blank: $\operatorname{arccot}'(x) = \underline{-\frac{1}{1+x^2}}$. cot^{-1} X + tan^{-1} X = \frac{\pi}{2}

- (iii) Find dy/dx , where $y = x^{\sqrt{x}}$. e^{ln x \cdot \sqrt{x}} \rightarrow x^{\sqrt{x}} \cdot (\frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}})

- (iv) Find $f'(x)$, where $f(x) = \int_{\cos(x)}^1 \sqrt{1-t^2} dt$. = \left(- \int_1^{\cos x} \sqrt{1-t^2} dt \right)' = -(\sqrt{1-\cos^2 x})(-\sin x) = \sin^2 x

- (v) Decompose the following fraction into partial fractions:

$$\frac{1}{(x-1)^2(x-2)}$$

$$\frac{1}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2} \quad \text{X=1: 1 = B(-1) \rightarrow B=-1}$$

$$1 = A(x-1)(x-2) + B(x-2) + C(x-1)^2$$

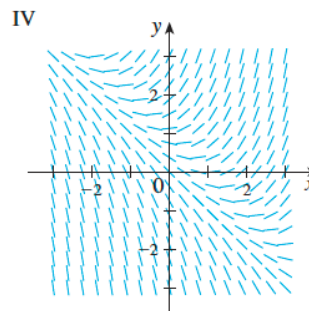
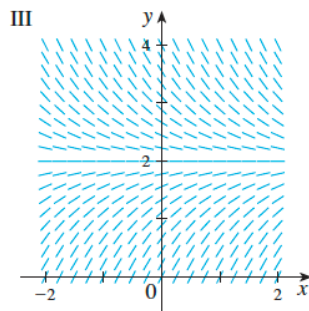
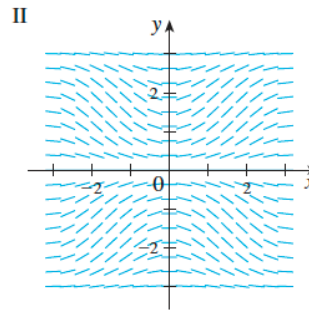
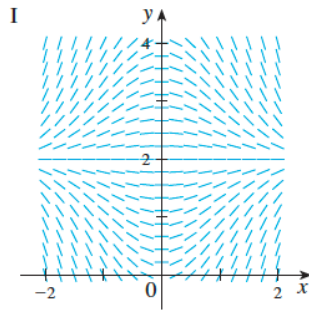
$$\text{X=1: } 1 = B(-1) \rightarrow B=-1$$

$$\text{X=2: } 1 = C(1) \rightarrow C=1$$

$$\text{X=3: } 1 = A(2) + (-1)(1) + 1 \cdot 2^2 \rightarrow A=-1$$

3. [4] Match the following differential equations (1) to (4) with their slope fields (I) to (IV). No justification is required

(1) $y' = \sin x \sin y$ (2) $y' = 2 - y$ (3) $y' = x(2 - y)$ (4) $y' = x + y - 1$



Slope = $\frac{1}{1+1^2} = \frac{1}{2}$

4. [4] Find an approximated value for $\arctan(1.01)$ by using standard linear approximation centred at $x = 1$. State your final answer in three decimal places (e.g., $e \approx 2.718$). = $\frac{\pi}{4} + 0.01 \cdot \frac{1}{2} = 0.786$

5. [6+3] Consider the integral

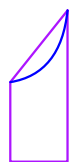
$$\int_0^{1.2} e^{x^2} dx$$

$T = \frac{0.2}{2} (e^0 + 2e^{0.04} + 2e^{0.16} + 2e^{0.36} + 2e^{0.64} + 2e^1 + e^{1.44})$
 $S = \frac{0.2}{3} (e^0 + 4e^{0.04} + 2e^{0.16} + 4e^{0.36} + 2e^{0.64} + 4e^1 + e^{1.44})$

and supposing we use the Trapezoidal Rule and Simpson's Rule to approximate it with the number of intervals $n = 6$.

(i) Write down the expressions of the two approximated values (T and S) in the form of sums. No simplification is required.

(ii) Will the Trapezoidal Rule over-estimate or under-estimate the true value of the integral? Briefly explain why.



$\frac{d}{dx} e^{x^2} = 2xe^{x^2} > 0 \quad \forall x \in [0, 1.2]$

Concave up $\rightarrow$ graph below lines connecting y_{i-1} & y_i

$$2X_1^3 - 3X_1^2 + 4 =$$

6. [3+3] Consider the function $f(x) = 2x^3 - 3x^2 + 4$ defined on $D = [1, \infty)$.

(i) Prove that f^{-1} exists by showing that f is injective (one-to-one).

(ii) Find $(f^{-1})'(4)$.
$$= \frac{1}{f'(\frac{3}{2})} = \frac{1}{6(\frac{3}{2})^2 - 6 \cdot \frac{3}{2}} = \frac{2}{9}$$

 $\because f(\frac{3}{2}) = 4$

7. [6] Find the area enclosed by the curves $y = 2x^2 + x - 9$ and $y = x - 1$.

8. [16] Evaluate the following limits, or explain why they do not exist.

(i) $\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot x} = e^{\lim_{x \rightarrow 0^+} \ln(1 + \sin(4x)) \cdot \cot x} = e^{\lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin(4x))}{\tan x}} = e^{\lim_{x \rightarrow 0^+} \frac{\frac{1}{1 + \sin(4x)} \cdot 4 \cos(4x)}{\sec^2 x}} = e^{\frac{1 \cdot 4}{1^2}} = e^4$

(ii) $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \dots + \frac{n}{n^2 + (n-1)^2} + \frac{n}{n^2 + n^2} \right)$

(iii) $\lim_{x \rightarrow \infty} \left(\frac{a^x - 1}{(a-1)x} \right)^{\frac{1}{x}}$, where $a \in (0, 1)$ is a constant.

(iv) $\lim_{x \rightarrow 0} \frac{x - \int_0^x e^{t^2} dt}{x^2 \sin(2x)} = \frac{1 - e^{x^2}}{2x \sin 2x + 2x^2 \cos 2x} = \frac{-2xe^{x^2}}{2 \sin 2x + 4x \cos 2x - 2x^2 \sin 2x} = \frac{-e^{x^2} - 2x^2 e^{x^2}}{2 \cos 2x + 4(\cos 2x - 2x \sin 2x) - 2(2x^2 \cos 2x + 2x \sin 2x)} = \frac{-1}{2 + 4(1) - 2(0)} = -\frac{1}{6}$

9. [24] Find the following integrals.

(i) $\int \frac{e^{1/x}}{x^2} dx = -\int \left(-\frac{1}{x^2}\right) e^{1/x} dx = -e^{1/x} + C$

(ii) $\int_0^{1/2} \frac{x^2}{\sqrt{1-x^2}} dx = \int_0^{\pi/6} \frac{\sin^2 \theta}{\cos \theta} \cos \theta d\theta = \int_0^{\pi/6} \frac{1 - \cos 2\theta}{2} d\theta = \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/6} = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$

(iii) $\int_0^1 \arccos x dx = \int_{\pi/2}^0 \theta (-\sin \theta d\theta) = \int_0^{\pi/2} \theta \sin \theta d\theta = [-\theta \cos \theta + \sin \theta]_0^{\pi/2} = (1) - (0) = 1$

(iv) $\int_{-3}^1 \frac{x+8}{x^2+6x+13} dx = \int_{-3}^1 \frac{1}{2} \left(\frac{2x+6}{x^2+6x+13} \right) dx + \int_{-3}^1 5 \cdot \frac{1}{(x+3)^2+4} dx$

(v) $\int_0^{\pi/4} \tan^5(x) dx = \int_0^{\pi/4} \tan^3(x) \tan(x) dx = \int_0^{\pi/4} \tan^2(x) \tan(x) dx = \int_0^{\pi/4} (\sec^2(x) - 1) \tan(x) dx = \left[\frac{\tan^4(x)}{4} - \frac{\tan^2(x)}{2} - \ln|\cos(x)| \right]_0^{\pi/4} = \frac{1}{4} - \frac{1}{2} + \ln \sqrt{2} = \ln \sqrt{2} - \frac{1}{4}$

(vi) $\int_0^\infty \frac{1}{1+x^4} dx$ (hint: consider $u = x - \frac{1}{x}$.)

$u = x - \frac{1}{x} \rightarrow du = \left(1 + \frac{1}{x^2}\right) dx = \frac{x^2+1}{x^2} dx$

$u = \frac{x^2-1}{x}$
 $u^2 = x^2 - 2 + \frac{1}{x^2}$
 $u^2 + 2 = \frac{x^4+1}{x^2}$

$\therefore \int_0^\infty \frac{1+x^2}{1+x^4} dx = \int_0^\infty \frac{x^2}{1+x^4} \cdot \frac{1+x^2}{x^2} dx = \int_0^\infty \frac{1}{u^2+2} \cdot du = \frac{1}{\sqrt{2}} \int_0^\infty \frac{1}{u^2+1} du = \frac{1}{\sqrt{2}} \left[\arctan u \right]_0^\infty = \frac{1}{\sqrt{2}} \cdot \frac{\pi}{2} = \frac{\pi}{2\sqrt{2}}$

10. [6] Find the solution to the initial value problem

$$xy' = y + x^2 \sin x, \quad x > 0, \quad y(\pi) = 0.$$

$v(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = x^{-1} = \frac{1}{x}$

$y' - \frac{1}{x}y = x \sin x$ $\frac{1}{x} \cdot y' - \frac{1}{x^2} \cdot y = \sin x$

$(\frac{1}{x} \cdot y)' = \sin x \rightarrow \frac{1}{x} \cdot y = -\cos x + C$ $y = x(-\cos x + C)$

11. [6] You have a certain area A of material to make a cylindrical can of radius r and height h . The material of area A includes the top, the bottom and the side of the cylinder. Find the values of r and h so that the volume of the cylinder is maximized.
- $A = 2\pi r(r+h)$ Find $\max[\pi r^2 h]$
- $r(r+h) = \frac{A}{2\pi}$ $V = \pi r^2 h = \pi r^2 (\frac{A}{2\pi r} - r) = \frac{Ar}{2} - \pi r^3$
- $0 = \frac{dV}{dr} = \frac{A}{2} - 3\pi r^2$ $(\frac{d^2V}{dr^2} = -6\pi r < 0) \rightarrow \frac{dV}{dr} = 0 \rightarrow \text{Maximum}$
- $3\pi r^2 = \frac{A}{2}$ $r^2 = \frac{A}{6\pi} \Rightarrow r = \sqrt{\frac{A}{6\pi}}, h = \frac{A}{2\pi \sqrt{\frac{A}{6\pi}}} - \sqrt{\frac{A}{6\pi}} = \sqrt{\frac{3A}{2\pi}} - \sqrt{\frac{A}{6\pi}} = \sqrt{\frac{2A}{3\pi}}$

12. [6] Find the length of the curve given by

$$y = \int_0^x \sqrt{\cos(2t)} dt, \quad 0 \leq x \leq \frac{\pi}{2}.$$

$$\frac{dy}{dx} = \sqrt{\cos 2x} \rightarrow L = \int_0^{\pi/2} \sqrt{1 + \cos 2x} dx = \int_0^{\pi/2} \sqrt{2} \cdot \cos x dx = \sqrt{2} [\sin x]_0^{\pi/2} = \sqrt{2}$$

13. [6+5+4] The curve given by

$$y = \arcsin\left(\frac{x}{a}\right), \quad 0 \leq x \leq a,$$

$$\sin y = \frac{x}{a} \rightarrow x = a \sin y$$

is revolved around the y -axis to form an open bowl facing up. Starting from time $t = 0$, water is let in from the top of the bowl at a volume rate of $r(t)$ given by

$$r(t) = Ae^{-\frac{t}{k}}, \quad t \geq 0.$$

Here, A , a , and k are all positive constants.

- (i) Find the volume of the bowl. $V = \int_0^{\pi/2} \pi (a \sin y)^2 dy = \pi a^2 \int_0^{\pi/2} \frac{1 - \cos 2y}{2} dy = \pi a^2 \left(\frac{y}{2} - \frac{\sin 2y}{4} \right)_0^{\pi/2} = \pi a^2 \left(\frac{\pi}{4} - 0 \right) = \frac{\pi^2 a^2}{4}$

- (ii) For a given pair of values of A and k , find the value of a so that the bowl will become full (but not overflow) as time t goes to infinity. Use this value of a in part (iii) below.

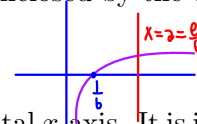
- (iii) Find the time point t_0 when the water level in the bowl is at $y = \pi/4$.
- Water volume = $W(t) = \int Ae^{-\frac{t}{k}} dt = [-kAe^{-\frac{t}{k}}]_0^t = 0 + kA = \frac{\pi^2 a^2}{4} \therefore a = \frac{2\sqrt{kA}}{\pi}$

$$V(t) = \int_0^{\pi/4} \pi (a \sin y)^2 dy = \pi a^2 \left(\frac{y}{2} - \frac{\sin 2y}{4} \right)_0^{\pi/4} = \pi a^2 \left(\frac{\pi}{8} - \frac{1}{4} \right) = \left[-\frac{\pi^2 a^2}{4} e^{-\frac{t}{k}} \right]_0^{t_0} = \frac{\pi^2 a^2}{4} - \frac{\pi^2 a^2}{4} e^{-\frac{t_0}{k}}$$

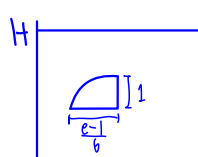
14. [2+5] Consider a plate formed by the region enclosed by the curves

$$y = \ln(bx), \quad b > 0,$$

$e^y = bx \rightarrow x = \frac{e^y}{b}$



the vertical line $x = a$, where $a > 0$, and the horizontal x -axis. It is immersed in water with a depth of H , i.e., the water surface is at $y = H$. The weight density of the water is w . Assume that $ab = e$, where e is Euler's number.



6

$ab > 1 \rightarrow a > \frac{1}{b}$

$l = \frac{e - e^y}{b}$

$e^{-\frac{t_0}{k}} = \frac{1}{2} + \frac{1}{\pi}$

$\frac{t_0}{k} = \ln\left(\frac{2\pi}{\pi+2}\right)$

$t_0 = k \ln\left(\frac{2\pi}{\pi+2}\right)$

- (i) Find the smallest value L of H so that the plate is totally immersed in the water. $L = 1 = \text{Height of plate}$
- (ii) Compute the force F exerted against one side of the plate when it is completely immersed in the water with the depth equal to the smallest value L above.

$$\int_0^1 W(1-y) \left(\frac{e-ey}{b} \right) dy = \frac{W}{b} \int_0^1 ye^y - e^y - ey + e \, dy = \frac{W}{b} \left[ye^y - 2e^y - \frac{ey^2}{2} + ey \right]_0^1 = \frac{W}{b} \left(2 - \frac{e}{2} \right)$$

15. [12] For each of the following improper integrals, determine whether it converges or diverges. $\because \frac{1}{x} < \frac{1}{e^x} \quad \forall x \geq 2$

(i) $\int_2^\infty \frac{e^{-x}}{x} dx < \int_2^\infty \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_2^\infty = \frac{1}{2}$ Converges

$\lim_{x \rightarrow 1} \frac{\frac{x}{x^2-1}}{\frac{2}{3(x^2-1)}} = \frac{3}{2} \lim_{x \rightarrow 1} \frac{x}{x^2-1} = \frac{3}{2} \lim_{x \rightarrow 1} \frac{x+1}{x^2+x+1}$ (ii) $\int_1^2 \frac{x}{x^3-1} dx =$ Diverges

$\int_1^2 \frac{1}{x^2-1} dx = \int_1^2 \frac{1}{2} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$ (iii) $\int_1^{\pi/2} \tan^2(\theta) d\theta$ $\because \lim_{\theta \rightarrow \pi/2} \frac{\tan^2 \theta}{\sec^2 \theta} = \lim_{\theta \rightarrow \pi/2} \sin^2 \theta = 1$

$\approx \frac{1}{2} [\ln|x-1| - \ln|x+1|]_1^2 = \frac{1}{2} (-\ln 3 - (\ln 0) + \ln 2) \rightarrow \int_0^{\pi/2} \tan^2 \theta d\theta$ and $\int_0^{\pi/2} \sec^2 \theta d\theta$ grows at the same rate $= [\tan \theta]_0^{\pi/2} = \infty$

Diverges